

SAMIR BHAUSAHEB AJAGEKAR

Location: Kolhapur, Maharashtra, India | **Email:** samirajagekar@gmail.com
Portfolio: <https://samir-ajagekar-portfolio.netlify.app> | **GitHub:** github.com/samirajagekar |
LinkedIn: linkedin.com/in/samirajagekar

PROFESSIONAL SUMMARY

Proactive and detail-oriented Computer Engineering professional with hands-on expertise in full-stack web development, AI-driven applications, and software testing. Skilled in modern JavaScript frameworks (React.js), backend architectures (FastAPI, Node.js), and database integrations. Proven ability to design and implement intelligent conversational systems and responsive web applications with clean, maintainable code.

TECHNICAL SKILLS

Category	Technologies & Tools
Frontend	React.js, JavaScript (ES6+), HTML5, CSS3, Bootstrap, Vite
Backend & AI	Python, FastAPI, Node.js, LangChain, Gemini API, Text-to-SQL RAG
Mobile & Database	Java, Kotlin (Android Development), SQL, Firebase, SQLite
Developer Tools	Git, GitHub, VS Code, Postman, Chrome DevTools, Netlify
Core Concepts	Data Structures & Algorithms, Software Testing (Unit, Black/White Box), Network Security

KEY PROJECTS

Text-to-SQL Retrieval-Augmented Generation (RAG) Chatbot

Tech Stack: Python, FastAPI, LangChain, Gemini Models, SQL

- Designed and implemented an intelligent conversational system translating natural language queries into executable SQL commands.
- Integrated LangChain SQL chains and schemas with Google's Gemini models for context-aware query generation and data retrieval.
- Architected secure API endpoints using FastAPI, streamlining natural language analytics and reporting pipelines.
- Constructed entity-relationship models, data flow diagrams (DFD), and thorough validation routines to prevent SQL injection vulnerabilities.

Smart Attendance System with QR Authentication

Tech Stack: Android (Kotlin/Java), Firebase Realtime Database, CameraX

- Developed a native mobile application facilitating touchless student attendance verification via dynamic QR code generation and scanning.
- Engineered real-time synchronization pipelines with Firebase to maintain tamper-proof, timestamped attendance records.
- Built an intuitive administrative dashboard interface providing instant attendance reporting and exportable analytics.

Personal Developer Portfolio Web Application

Tech Stack: React.js, JavaScript (ES6+), Bootstrap, Netlify

- Constructed and deployed a highly responsive, modern portfolio website showcasing completed technical projects and proficiencies.
- Optimized build performance and bundling efficiency by structuring the application using Vite.
- Configured continuous integration and automated deployment workflows through Git and Netlify.

EDUCATION

Diploma in Computer Engineering

Sant Gajanan Maharaj Rural Polytechnic (MSBTE) | Mahagaon, Kolhapur, Maharashtra

- **Graduation:** 2026
- **Key Coursework:** Data Structures & Algorithms, Software Testing, Relational Database Management Systems, Network & Information Security, Operating Systems.

ACADEMIC ACHIEVEMENTS & ACTIVITIES

- **Algorithmic Problem Solving:** Actively practicing foundational data structures and algorithms focusing on arrays, hashing, and dynamic programming patterns.
- **Software Quality Assurance:** Experienced in creating comprehensive defect reports, execution test plans, and automated verification suites following standard testing lifecycles.